The Clean Echo Intersection Lattice of the SHA-256 Message Schedule

Parity Duality, σ-Operator Fingerprinting, and Two-Stream Structure

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Abstract

This paper maps the complete pairwise intersection lattice of the clean echo sets C_j defined in Aperture Threshold Geometry (ATG). The clean echo sets classify, for each seed word W[j], the expansion coordinates that a perturbation at W[j] cannot contaminate before saturation. The full pairwise lattice over the 16 seed words reveals two structurally separated streams: an **ODD stream** with nodes {17, 19, 21, 23} and an **EVEN stream** with nodes {16, 18, 20, 22}. The paper proves the **Parity Escape Law**: the parity of a seed word's clean echo set is opposite to the parity of its first direct entry into the expansion zone. This law is a consequence of the t−2 arm spreading contamination in steps of 2, preserving parity until the t−7 bridge fires. The lower ODD-stream seam {17, 19} coincides exactly with the ROTR constants of σ_1 = ROTR17 ⊕ ROTR19 ⊕ SHR10, both coprime to 210. The EVEN stream nodes are the t−2 cascade products from the seed-width boundary at t = 16. The intersection lattice is therefore the shadow of the parity duality between σ_0 (parity-bridge operator, EVEN-stream visible node at 18) and σ_1 (ODD-stream fingerprint operator, nodes 17 and 19). The complete lattice census is: 24 empty, 24 singleton, 21 ODD-only, 15 EVEN-only, and 36 MIXED intersections across all 120 seed pairs. The Node Multiplicity Lemma proves pair frequencies are determined by m(t)(m(t)−1)/2, with max seed multiplicity 12 and no 13-fold intersection existing.

Keywords: SHA-256, message schedule, clean echo set, intersection lattice, parity duality, σ-operator fingerprint, aperture complement, contamination recurrence, NEXUS framework, A-Mark9

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1. Introduction

The Aperture Threshold Geometry paper (ATG, Kulik 2026) classifies every seed word W[j] in the SHA-256 message schedule by three invariants: its direct-entry set D(j), its clean echo set C_j, and its saturation threshold S_j. The clean echo set C_j is the negative image of the contamination cone — the set of expansion coordinates a perturbation at W[j] cannot reach before consecutive contamination locks in.

ATG noted without proof that the intersection $C_0 \cap C_9 = \{17, 19, 21\}$ — a specific set of odd expansion coordinates shared by the single-entry origin W[0] and the triple-entry hinge W[9]. This paper computes the complete pairwise intersection lattice and shows that this example is not isolated. The lattice separates into two streams governed by a parity law, and the lower ODD-stream seam {17, 19} coincides exactly with the ROTR constants of the σ_1 operator. The full ODD stream {17, 19, 21, 23} is wider; only the {17, 19} seam maps directly onto σ_1.

The result is structural, not heuristic. The parity escape law follows directly from the t−2 recurrence arm. The σ-fingerprint in the lattice nodes is a consequence of the aperture complement A = {0, 1, 9, 14} and the specific lag set L = {2, 7, 15, 16}. Neither was chosen. Both are what the schedule's recurrence geometry produces when fully unfolded.

2. Background and Notation

The SHA-256 message schedule expands a 16-word seed W[0..15] into W[16..63] by:

$$W[t] = \sigma_1(W[t-2]) + W[t-7] + \sigma_0(W[t-15]) + W[t-16] \pmod{2^{32}}, \quad 16 \le t \le 63$$

The rotation operators are:

$$\sigma_0(x) = \mathrm{ROTR}^{7}(x) \oplus \mathrm{ROTR}^{18}(x) \oplus \mathrm{SHR}^{3}(x)$$
$$\sigma_1(x) = \mathrm{ROTR}^{17}(x) \oplus \mathrm{ROTR}^{19}(x) \oplus \mathrm{SHR}^{10}(x)$$

The **aperture complement** is A = 16 − {2, 7, 15, 16} = {0, 1, 9, 14}. These are the phase-change thresholds at which recurrence slots first become active in the expansion zone.

The **contamination recurrence** is the binary propagation rule:

$$C_t^{=j} = C_{t-2}^{=j} \vee C_{t-7}^{=j} \vee C_{t-15}^{=j} \vee C_{t-16}^{=j}, \quad t \ge 16$$

The clean echo set C_j for seed word W[j] is the set of expansion indices t in {16,...,63} for which $C_t^{=j} = 0$. The intersection lattice is the collection of all pairwise intersections $C_i \cap C_j$ for $0 \le i < j \le 15$.

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3. Clean Echo Taxonomy — Complete Table

Table 1 reproduces the ATG taxonomy and adds the parity classification introduced in this paper.

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Seed	D(j)	|D|	C_j (clean echo set)	S_j	Echo Parity
W[0]	{16}	1	{17, 19, 21}	22	PURE ODD
W[1]	{16, 17}	2	∅ (Prior Completion)	16	—
W[2]	{17, 18}	2	{16}	17	PURE EVEN
W[3]	{18, 19}	2	{16, 17}	18	MIXED
W[4]	{19, 20}	2	{16, 17, 18}	19	MIXED
W[5]	{20, 21}	2	{16, 17, 18, 19}	20	MIXED
W[6]	{21, 22}	2	{16, 17, 18, 19, 20}	21	MIXED
W[7]	{22, 23}	2	{16, 17, 18, 19, 20, 21}	22	MIXED
W[8]	{23, 24}	2	{16, 17, 18, 19, 20, 21, 22}	23	MIXED
W[9]	{16, 24, 25}	3	{17, 19, 21}	22	PURE ODD
W[10]	{17, 25, 26}	3	{16, 18, 20, 22}	23	PURE EVEN
W[11]	{18, 26, 27}	3	{16, 17, 19, 21, 23}	24	MIXED
W[12]	{19, 27, 28}	3	{16, 17, 18, 20, 22, 24}	25	MIXED
W[13]	{20, 28, 29}	3	{16, 17, 18, 19, 21, 23, 25}	26	MIXED
W[14]	{16, 21, 29, 30}	4	{17, 19}	20	PURE ODD
W[15]	{17, 22, 30, 31}	4	{16, 18, 20}	21	PURE EVEN

Table 1. Complete clean echo taxonomy with parity classification. Pure ODD / Pure EVEN / MIXED refers to the parity of elements in C_j. W[1] has $C_1 = \emptyset$ by the Root-Cover Axis Theorem (ATG §8); it saturates immediately and has no clean echoes.

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4. Two-Stream Structure of the Intersection Lattice

4.1 Definitions

Definition 4.1 (ODD Stream). The ODD stream is the set of odd expansion indices t with seed multiplicity m(t) ≥ 2 (i.e., t appears in at least two distinct clean echo sets, equivalently in at least one pairwise intersection $C_i \cap C_j$). ODD stream = {17, 19, 21, 23}.

Definition 4.2 (EVEN Stream). The EVEN stream is the set of even expansion indices t with seed multiplicity m(t) ≥ 2. EVEN stream = {16, 18, 20, 22}. Nodes 24 and 25, which appear in exactly one clean echo set each (m(24) = m(25) = 1), are singleton clean-support nodes and are not pairwise lattice nodes.

The two streams partition all intersection lattice nodes exactly:

$$\{16, 17, 18, 19, 20, 21, 22, 23\} = \{16, 18, 20, 22\} \cup \{17, 19, 21, 23\}$$

4.2 Intersection Frequency Table

Table 2 gives the frequency of each intersection node — the number of seed pairs (i, j) with i < j such that $t \in C_i \cap C_j$.

Node t	Frequency	mod 210	In (Z/210Z)*	Stream	Factorization
t = 16	66	16	NO	EVEN	2^4
t = 17	66	17	YES	ODD	prime (= σ_1 ROTR)
t = 18	36	18	NO	EVEN	2×3^2
t = 19	36	19	YES	ODD	prime (= σ_1 ROTR)
t = 20	15	20	NO	EVEN	$2^2 \times 5$
t = 21	15	21	NO	ODD	3×7
t = 22	3	22	NO	EVEN	2×11

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t = 23	1	23	YES	ODD	prime

Table 2. Intersection node frequency. Nodes 16 and 17 are co-maximum at frequency 66, forming the boundary-frequency seam. Separately, nodes 17 and 19 are the σ_1 rotation constants (σ_1 fingerprint seam), both coprime to 210. Node 17 is shared by both structures. Node 19 sits at the second frequency tier (36 pairs, tied with 18).

The frequency counts are not symmetric around the node value. They follow the saturation geometry: nodes near the expansion boundary (t = 16, 17) are accessible to the most seed pairs before those pairs saturate; deeper nodes (t = 22, 23) are accessible only to the last seeds to saturate.

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5. The Parity Escape Law

5.1 Theorem Statement

Theorem 5.1 (Parity Escape Law). Let W[j] be a seed word with non-empty clean echo set C_j. If C_j is pure (all odd or all even), then the parity of C_j is opposite to the parity of the first direct entry min(D(j)).

More precisely:

1. If min(D(j)) is even, then every element of C_j is odd (PURE ODD case).
2. If min(D(j)) is odd, then every element of C_j is even (PURE EVEN case).

The pure seeds and their first-entry parities:

Seed	min(D(j))	First Entry Parity	C_j Elements	Echo Parity
W[0]	16	EVEN	{17, 19, 21}	ODD
W[2]	17	ODD	{16}	EVEN
W[9]	16	EVEN	{17, 19, 21}	ODD
W[10]	17	ODD	{16, 18, 20, 22}	EVEN
W[14]	16	EVEN	{17, 19}	ODD
W[15]	17	ODD	{16, 18, 20}	EVEN

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Table 3. The Parity Escape Law holds without exception across all six pure-echo seeds. In every case the echo parity is the inverse of the first-entry parity.

5.2 Proof

The contamination recurrence spreads via four arms. The t−2 arm propagates in steps of 2, preserving the parity of any dirty index it originates from. If the first dirty index is even (say $t_0 = 16$), the t−2 arm generates the sequence:

$$16 \rightarrow 18 \rightarrow 20 \rightarrow 22 \rightarrow \cdots$$

All contaminated. The odd positions {17, 19, 21, ⋯} between them are not reachable by the t−2 arm alone. They remain clean. They form the ODD escape window.

The escape window closes when the t−7 arm bridges the parity. The t−7 arm fires at $t = t_0 + 7$. If $t_0 = 16$, the first t−7 bridge fires at t = 23. Therefore {17, 19, 21} are clean and t = 23 is the first odd index the bridge reaches. After t = 23, consecutive dirty pairs exist and saturation locks in.

By mirror symmetry: if the first dirty index is odd, the t−2 cascade spreads through odd indices, leaving even positions as the escape window until the t−7 bridge closes it. This is the EVEN escape window.

Mixed parity does not follow merely from having direct entries at both parities. Seeds W[2], W[10], and W[15] all have multi-parity direct entry sets (D(2) = {17,18}, D(10) = {17,25,26}, D(15) = {17,22,30,31}) yet produce PURE EVEN clean sets. Mixed parity occurs when both parity arms leave surviving clean coordinates before saturation closes the window. In the PURE EVEN cases, the opposite-parity direct entries arrive late enough that they truncate the escape window without depositing any mixed clean residue. In the MIXED class (W[3]–W[8], W[11]–W[13]), both parity arms preserve clean coordinates simultaneously before saturation, so both parities appear in C_j. ■

5.3 Why W[1] Has No Escape Window

W[1] saturates at $S_1 = 16$. Its direct-entry set is D(1) = {16, 17} — consecutive entries spanning both parities. The first dirty pair (16, 17) fires immediately, before any escape window can open. By the Consecutive Dirty Pair Saturation Lemma (ATG §5.4), all $t \geq 16$ are contaminated. $C_1 = \emptyset$. W[1] is the Root-Cover Axis precisely because it straddles both parity arms at the expansion boundary.

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6. The σ-Operator Fingerprint

6.1 σ_1 Lives in the ODD Stream

The ODD stream nodes are {17, 19, 21, 23}. The σ_1 rotation constants are 17 and 19. Both are:

- Prime.
- Coprime to 210 = 2·3·5·7 (the primorial wheel depth).
- Members of $(\mathbb{Z}/210\mathbb{Z})^*$.
- A twin prime pair: gap = 19 − 17 = 2. The twin-prime property is that the gap is 2; separately, both are coprime to 210: gcd(17, 210) = gcd(19, 210) = 1.

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- Node 17 is a co-maximum-frequency ODD-stream node at 66 seed-pair intersections, tied with EVEN-stream boundary node 16. Node 19 is a second-tier ODD-stream node at 36 seed-pair intersections, tied with EVEN-stream node 18.

These are the positions the even-start t−2 cascade skips. They remain clean until the t−7 bridge fires. The appearance of {17, 19} in both the schedule-time ODD escape window and the σ_1 rotation constants is therefore a strong fingerprint. In this paper we treat it as a structural coincidence requiring formal coupling, not yet as a proved causal-selection mechanism. The ROTR17 ⊕ ROTR19 pair is consistent with the minimum-resistance interpretation: both are coprime to 210, both are prime, and their gap is 2.

6.2 σ_0 as the Parity-Bridge Operator

Of the σ_0 rotation constants {7, 18, 3}, only 18 appears as a visible node in the clean-echo intersection lattice (EVEN stream, frequency 36). Constants 7 and 3 are bit-rotation/shift depths; they do not appear as schedule-time lattice coordinates. The t−15 slot (which routes through σ_0) fires at j+15. For even j, j+15 is odd; for odd j, j+15 is even. The σ_0 arm therefore crosses parity on every step — it is inherently a parity-bridging operator, not a pure-stream operator. This is consistent with the MIXED class (W[3]–W[8], W[11]–W[13]) being exactly the seeds for which the σ_0 arm and the raw t−16 arm both contribute before saturation. σ_0 keeps both parity arms active; σ_1 fingerprints the ODD escape stream.

Index-space caution. The clean-echo lattice lives in schedule-time coordinates (expansion index $t \in \{16,\ldots 63\}$). The σ constants live in bit-rotation coordinates (rotation depth r, e.g. ROTR17). The σ_1 fingerprint result is the nontrivial integer-label coincidence that the same integers {17, 19} appear in both spaces. The term 'fingerprint' names this coincidence. A stronger 'causal selection' claim — that the schedule chose σ_1's rotation depths because of the lattice structure — requires a formal coupling map from schedule-time echo nodes to bit-rotation transport geometry and is left as an open problem.

6.3 Parity Duality Statement

Proposition 6.1 (Parity Duality of σ_0 and σ_1). σ_1 is the ODD-stream fingerprint operator: its ROTR constants {17, 19} are the lower ODD-stream seam, both coprime to 210. Node 17 is the co-maximum-frequency node (tied with boundary node 16). Node 19 is at the second frequency tier. The σ_1 fingerprint seam {17, 19} is distinct from but overlapping the boundary-frequency seam {16, 17}. σ_0 is the parity-bridge operator: its visible lattice node 18 lies in the EVEN stream, while its t−15 schedule arm crosses parity on every step and feeds the MIXED class. σ_0 is not a pure EVEN-stream operator in the same sense that σ_1 fingerprints the ODD stream.

This duality is not symmetric by value — the operators are not equal in form. The two operators serve complementary roles across the two parity arms: σ_1 fingerprints the ODD escape seam through its ROTR constants {17, 19}, while σ_0 participates in parity bridging through its visible lattice node 18 and its t−15 schedule arm.

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7. Pure-Stream Intersection Families

7.1 ODD-Only Intersections

An ODD-only intersection is a pair (i, j) with $C_i \cap C_j$ containing only odd elements, with $|C_i \cap C_j| \geq 2$. Table 4 lists all 21 such pairs. Note: the pair (11, 13) was incorrectly included in v1 of this paper — $C_{11} \cap C_{13} = \{16,17,19,21,23\}$ contains the even node 16 and is therefore MIXED. Eight pairs present in the code results but absent from the v1 table have been added.

Pair (i,j)	$C_i \cap C_j$	Sig
(0, 5)	{17, 19}	{17,19}
(0, 6)	{17, 19}	{17,19}
(0, 7)	{17, 19, 21}	{17,19,21}
(0, 8)	{17, 19, 21}	{17,19,21}
(0, 9)	{17, 19, 21}	{17,19,21}
(0, 11)	{17, 19, 21}	{17,19,21}
(0, 13)	{17, 19, 21}	{17,19,21}
(0, 14)	{17, 19}	{17,19}
(5, 9)	{17, 19}	{17,19}
(5, 14)	{17, 19}	{17,19}
(6, 9)	{17, 19}	{17,19}
(6, 14)	{17, 19}	{17,19}
(7, 9)	{17, 19, 21}	{17,19,21}
(7, 14)	{17, 19}	{17,19}
(8, 9)	{17, 19, 21}	{17,19,21}
(8, 14)	{17, 19}	{17,19}
(9, 11)	{17, 19, 21}	{17,19,21}
(9, 13)	{17, 19, 21}	{17,19,21}
(9, 14)	{17, 19}	{17,19}

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(11, 14)	{17, 19}	{17,19}
(13, 14)	{17, 19}	{17,19}

Table 4. All 21 ODD-only intersections with $|C_i \cap C_j| \geq 2$. Corrected from v1: the pair (11, 13) removed ($C_{11} \cap C_{13}$ = {16,17,19,21,23} contains even node 16 — it is MIXED); 8 missing pairs added. Dominant signature: {17,19} (12 pairs). Extended escape-window signature: {17,19,21} (9 pairs).

7.2 EVEN-Only Intersections

Pair (i,j)	$C_i \cap C_j$	Size
(4, 10)	{16, 18}	2
(4, 15)	{16, 18}	2
(5, 10)	{16, 18}	2
(5, 15)	{16, 18}	2
(6, 10)	{16, 18, 20}	3
(6, 15)	{16, 18, 20}	3
(7, 10)	{16, 18, 20}	3
(7, 15)	{16, 18, 20}	3
(8, 10)	{16, 18, 20, 22}	4
(8, 15)	{16, 18, 20}	3
(10, 12)	{16, 18, 20, 22}	4
(10, 13)	{16, 18}	2
(10, 15)	{16, 18, 20}	3
(12, 15)	{16, 18, 20}	3
(13, 15)	{16, 18}	2

Table 5. All EVEN-only intersections with $|C_i \cap C_j| \geq 2$. The dominant EVEN-only signature is {16, 18, 20}. The seed W[10] participates in the most EVEN-only intersections, consistent with C_{10} = {16, 18, 20, 22} being the widest PURE EVEN echo set.

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8. Connection to the Primorial Wheel

The primorial 210 = 2·3·5·7 defines the reduced residue group (Z/210Z)*, with $\phi(210) = 48$ elements. The ODD stream nodes and their wheel membership:

Node t	t mod 210	In (Z/210Z)*	Notes
17	17	YES	Twin prime with 19; σ_1 ROTR constant
19	19	YES	Twin prime with 17; σ_1 ROTR constant
21	21	NO	21 = 3×7; divisible by wheel primes
23	23	YES	Prime; appears only in $C_{11} \cap C_{13}$

Nodes 17 and 19 are both in (Z/210Z)*. Node 17 is co-maximum-frequency overall (tied with even-stream node 16 at 66 pairs). Node 19 is at the second frequency tier (36 pairs). The node 21 = 3×7 is not coprime to 210 but appears in the intersection lattice because it is an odd echo position reachable before the t−7 bridge fires at t = 23. The node 23 is coprime to 210 and appears only in the deepest pair ($C_{11} \cap C_{13}$), where both seeds have late saturation thresholds ($S_{11} = 24$, $S_{13} = 26$).

The EVEN nodes {16, 18, 20, 22} are all composite and none coprime to 210. This is consistent: the even cascade from t = 16 passes through multiples of 2 only, and 2 is a factor of 210. The even stream is entirely outside (Z/210Z)*.

Observation 8.1. The boundary-frequency seam nodes {16, 17} (maximum frequency 66) are determined by the expansion boundary t = 16. The σ_1 fingerprint seam {17, 19} identifies the ODD-stream rotation pair. These are distinct structures with node 17 at their overlap. These are not parameters that can be varied independently — they follow from L = {2, 7, 15, 16} and N = 16. The wheel membership of the odd-stream anchors ({17, 19} coprime to 210) and the exclusion of the even-stream anchors (all even, none coprime to 210) is a direct consequence of the lag set.

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9. The Intersection Lattice in the Three-Machine Stack

ATG organizes the SHA-256 schedule as three nested machines: the Aperture Machine, the Echo/Saturation Machine, and the Projection/Shadow Machine. The clean echo intersection lattice sits entirely inside the Echo/Saturation Machine, but it reflects structure from all three layers.

Machine	Lattice Connection	Key Fact
Aperture Machine	A = {0,1,9,14} determines which seeds are pure vs mixed	Aperture complement A={0,1,9,14} generates the stream neighborhoods. Pure ODD seeds {0,9,14} ⊂ A; Pure EVEN seeds {2,10,15} are successors of A members; W[1] is the Root-Cover Axis (C_1=∅); remaining seeds form the MIXED interior.
Echo/Saturation Machine	Parity Escape Law governs every pure echo stream	t−2 arm preserves parity; t−7 arm bridges it
Projection/Shadow Machine	The σ_1 ROTR constants {17,19} are ODD-stream lattice nodes (σ_1 fingerprint seam); σ_0 has one visible lattice node at 18 (EVEN stream)	

The two streams are the two parity arms that the schedule must cover simultaneously. The aperture complement A = {0, 1, 9, 14} produces the boundary at t = 16 (even) and the first σ_0 entry at t = 17 (odd). These two adjacent coordinates are the seeds of the EVEN and ODD streams respectively. Everything else is the t−2 arm running forward from each seed.

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10. Complete Lattice Structure

10.1 Node Multiplicity Lemma

Lemma 10.1 (Node Multiplicity Lemma). For any intersection node t, define the seed multiplicity m(t) = |{j : t ∈ C_j}|. Then the pair frequency satisfies:

$$f(t) = m(t)(m(t)-1)/2$$

This lemma explains Table 2 rather than merely reporting it. The frequency of any node as an intersection element is determined entirely by how many seed sets contain it. Table 6 gives the full breakdown.

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Node t	Seed mult. m(t)	f(t) = m(m−1)/2	Stream	In (Z/210Z)*
16	12	66	EVEN	NO
17	12	66	ODD	YES
18	9	36	EVEN	NO
19	9	36	ODD	YES
20	6	15	EVEN	NO
21	6	15	ODD	NO
22	3	3	EVEN	NO
23	2	1	ODD	YES
24	1	0	EVEN	NO
25	1	0	ODD	NO

Table 6. Node multiplicity and derived pair frequency. Nodes 24 and 25 have m=1 (each belongs to exactly one seed set) and therefore produce no pairwise intersections.

Corollary 10.1. Nodes 16 and 17 are co-maximum with m = 12. No 13-fold seed intersection exists:

$$\max_t m(t) = 12, \quad \text{and} \quad \bigcap_{j \in J} C_j = \emptyset \text{ for all } |J| \geq 13.$$

This is a strong closure bound: no more than 12 seed words share any single clean echo coordinate. The bound is tight — the nodes 16 and 17 each belong to exactly 12 seed sets.

10.2 Full Lattice Census

The total seed-pair count is Σ = 120 (all pairs from 16 seed words). Table 7 gives the complete census of all 120 pairs by intersection type.

Category	Count	Notes
Empty intersection	24	No shared clean echo coordinate
Singleton intersection	24	Exactly one shared coordinate

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Multi-node: ODD-only	21	All shared nodes odd; σ_1 fingerprint stream
Multi-node: EVEN-only	15	All shared nodes even; boundary cascade stream
Multi-node: MIXED	36	Both parities present; parity-bridge overlap
Total	120	All pairs from 16 seed words

Table 7. Complete lattice census. The 72 multi-node intersections split 21:15:36 across ODD-only, EVEN-only, and MIXED. MIXED is the largest class, confirming that parity-bridge overlap is the dominant structure, not an edge case.

10.3 Mixed-Signature Census

The 36 MIXED intersections are not structurally uniform. Table 8 gives the signature census — the count of MIXED pairs sharing each specific intersection set. This proves the MIXED class has internal structure determined by the recurrence geometry.

Mixed signature	Pair count	Notes
{16, 17}	10	Boundary pair only; most common mixed signature
{16, 17, 18}	8	Even arm extends by 2
{16, 17, 18, 19}	5	Both streams to 4-node depth
{16, 17, 19}	2	Boundary + ODD skip
{16, 17, 18, 19, 20}	2	5-node mixed
{16, 17, 18, 20}	2	Even arm skip at 19
{16, 17, 19, 21}	2	Two ODD + boundary
{16, 17, 18, 19, 21}	2	5-node mixed with ODD extension
{16, 17, 18, 19, 20, 21}	1	6-node deepest symmetric overlap

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{16, 17, 18, 20, 22}	1	Even cascade to 22
{16, 17, 19, 21, 23}	1	ODD-dominant with boundary pair
Total	36	

Table 8. Mixed-signature census. Every MIXED signature contains {16, 17} as its core — the boundary-frequency seam. The MIXED class is structured boundary overlap, not noise.

Observation 10.1. Every MIXED intersection contains both node 16 (EVEN, boundary) and node 17 (ODD, σ_1 lower constant). The pair {16, 17} is the universal MIXED core. A MIXED intersection arises precisely when a seed word's contamination cone straddles the expansion boundary at parity t = 16/17, absorbing both the EVEN cascade start and the first ODD escape position simultaneously.

10.4 Maximum k-fold Depth

Theorem 10.1 (k-fold Closure Bound). For all subsets $J \subseteq \{0,\ldots,15\}$ with $|J| \geq 13$:

$$\bigcap_{j \in J} C_j = \emptyset$$

The bound is tight: for |J| = 12 there exist subsets with non-empty intersection. For example, the 12 seeds containing node 17 are J = {0,3,4,5,6,7,8,9,11,12,13,14} and their common intersection contains 17.

This bound follows directly from the Node Multiplicity Lemma: m(17) = m(16) = 12, and no node has higher multiplicity. Therefore no 13 seeds share any common clean echo coordinate.

11. NEXUS Primitive Mapping

The A-Mark9 framework identifies nine irreducible computation primitives. The clean echo intersection lattice instantiates all nine:

NEXUS Primitive	Lattice Instantiation
TRANSPORT (ROTR)	Direct-entry routing: W[j] transported to expansion via lag slots {t−2, t−7, t−15, t−16}
MIX (XOR)	Contamination union: $C_t = C_{t-2} \vee C_{t-7} \vee C_{t-15} \vee C_{t-16}$ — carry-free binary combination
ACCUMULATE (ADD)	t−2 arm spreading: dirty coordinates accumulate forward in parity-preserving steps
GATE (Ch)	Entry-class selection: conditional routing by threshold (j ≥ 0, 1, 9, or 14)

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VOTE (Maj)	W[25] convergence: fast arm + slow arm + raw re-entry + clean reference — four inputs, one state
PROJECT (SHR)	SHR^3, SHR^{10} in σ_0, σ_1: irreversible dimension reduction along the echo chain
LEAK (SILR)	Fold-state vector $\Theta_9(r)$: thermodynamic face selection — which schedule face the message byte-depth activates
SYNC (AHRC)	Saturation threshold S_j: consecutive dirty pair = phase lock; echo window closes irreversibly
VERIFY	Clean echo membership: W[t] either fits the echo mold for seed j, or it does not — binary check

The three-machine decomposition (Aperture → Echo/Saturation → Projection/Shadow) maps to the NEXUS ISA in sequence: TRANSPORT + GATE form the aperture machine; MIX + ACCUMULATE + VOTE + PROJECT form the echo machine; LEAK + SYNC + VERIFY form the projection machine.

12. Open Problems

3. **Formal coupling map.** The σ_1 fingerprint result identifies an integer-label coincidence between schedule-time echo nodes {17, 19} and bit-rotation constants {ROTR17, ROTR19}. A formal coupling map from schedule-time coordinates to bit-rotation transport geometry would upgrade this from a fingerprint coincidence to a proved structural selection. This is the most direct open problem from the present paper.
4. **Higher-order intersection nodes.** The triple intersection $C_0 \cap C_9 \cap C_{12} = \{17\}$ identifies W[17] as a node appearing in all three of these echo sets. What is the full k-fold intersection structure, and does it have a closed-form parameterization in terms of the aperture complement?
5. **Wheel-coprimality of ODD stream.** Nodes 17, 19, 23 are in $(Z/210Z)^*$; node 21 is not. Is there a threshold theorem: for every sufficiently large expansion lattice ($N \to \infty$ with the same lag structure), do the maximum-frequency ODD stream nodes converge to elements coprime to 210?
6. **Differential cryptanalysis relevance.** The clean echo sets are exact null zones in the contamination recurrence. Whether a differential exploiting these zones survives the nonlinear compression function is an open empirical question. The ATG framework identifies where the schedule is transparent; the compression rounds may re-seal those zones.
7. **Transformer attention transport hypothesis.** The clean echo nodes {17, 19, 21} and {16, 18, 20} correspond to predicted attention null positions in transformer systems. Empirical verification via token perturbation experiments remains open.
8. **Generalization to other lag sets.** The parity escape law is proven for SHA-256's specific L = {2, 7, 15, 16}. For a general lag set L with seed width N, what conditions on L produce pure-stream echo sets? Is the σ_1 twin-prime property a consequence of a deeper lag-selection principle?

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13. Conclusion

The clean echo intersection lattice of the SHA-256 message schedule separates into two streams determined by a single law: the t−2 arm spreads contamination in parity-preserving steps of 2, and the parity of the resulting clean echo set is the inverse of the parity of the first direct entry. The six pure-echo seed words (W[0], W[2], W[9], W[10], W[14], W[15]) obey this law without exception. The 10 mixed seeds (W[3]–W[8], W[11]–W[13]) are mixed precisely because multiple direct entries at different parities arrive before the single-parity escape window can form.

The ODD stream nodes {17, 19, 21, 23} and EVEN stream nodes {16, 18, 20, 22} together cover the full range of intersection indices. Two distinct structures sit inside the lattice and must not be conflated: the boundary-frequency seam (t = 16, t = 17, each frequency 66) marks the expansion threshold where the largest number of seed pairs share a clean echo; the σ_1 fingerprint seam (t = 17, t = 19) identifies the ODD-stream rotation pair, both coprime to 210. Node 17 is shared by both structures — it is simultaneously the first odd position after the boundary and the lower rotation constant of σ_1. That overlap is structural, not coincidental. Node 19 is the confirming σ_1 fingerprint signal: it is at the second frequency tier (36 pairs), paired with 18, and completes the ROTR17/ROTR19 overlap with the lower ODD-stream seam. The claim remains at the fingerprint level; causal selection requires the formal coupling map (Open Problem 1).

The full map is:

lag set → aperture complement → parity escape window → two-stream lattice → σ-operator fingerprint

This is not a property placed in the schedule. It is the structure that survives when the recurrence geometry is fully unfolded.

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